

WORK MEASUREMENT / TIME STUDY

➤ *The application of techniques designed to establish the time for a qualified worker to carry out a specified job at a defined level of performance*

➤ *Work measurement refers to the estimation of standard time for an activity, that is the time allowed for completing one piece of job by using the prescribed method. Standard time can be defined as the time taken by an average experienced worker for the job with provisions for delays beyond the worker's control.*

WORK MEASUREMENT OBJECTIVES

- *Comparing alternative methods*
- *Assessing the correct initial manning*
- *Realistic costing*
- *Delivery date of goods*
- *Cost reduction & cost control*
- *Training new employees*
- *Find ineffective time in a process*
- *Evaluate worker's performance*
- *Facilitate operations scheduling*
- *Establish wage incentive schemes*

TECHNIQUES OF WORK MEASUREMENT

1. **Stop watch time study**(It deals with time and rate of working)
2. **Work sampling**(Large number of observations from random samples)
3. **Pre determined time standards (PTS)**(Basic human motion for increase in performance)
4. **Standard data.**(Time for a job in countineous process)

WORK MEASUREMENT TECHNIQUES

➤ *For repetitive work (short work cycle) or non-repetitive work;*

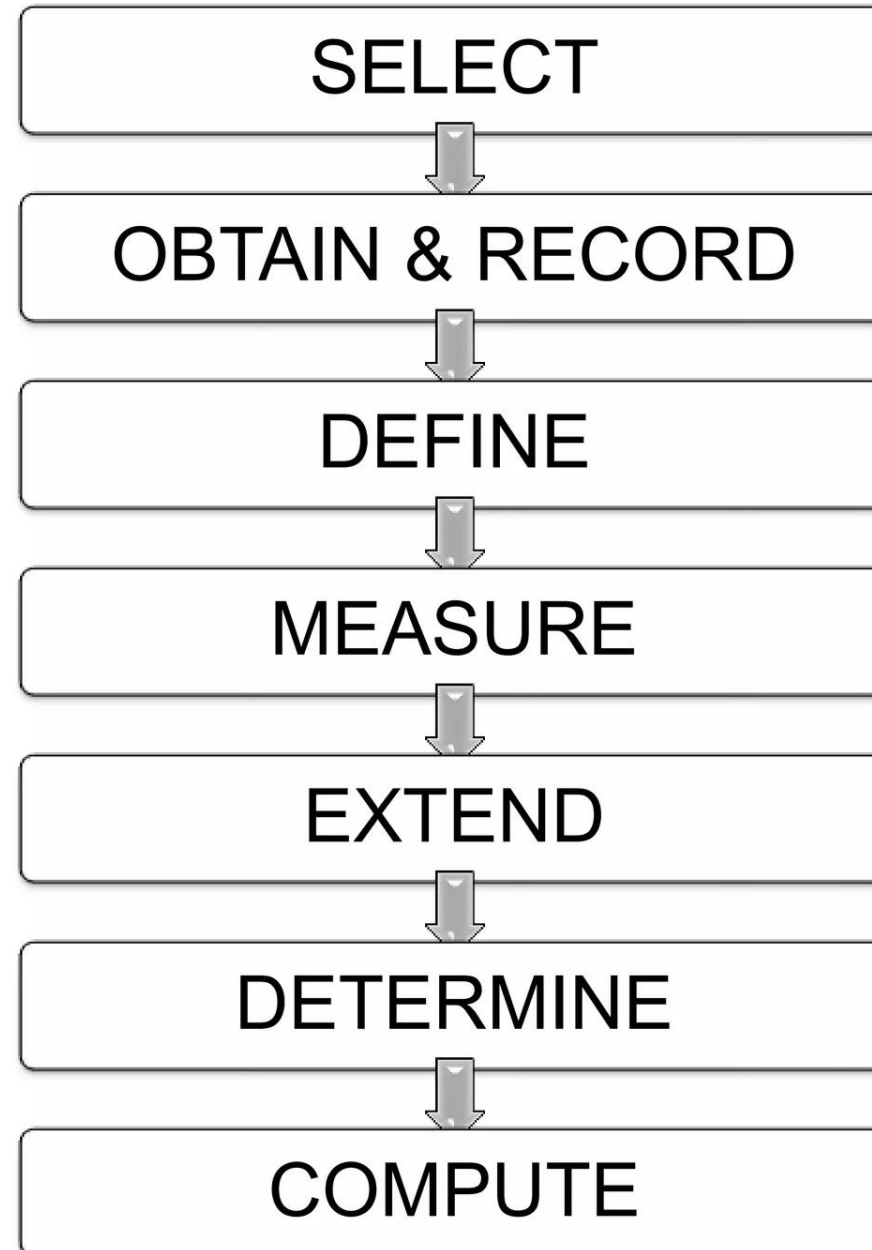
- ✓ *Time study (stop watch technique)*
- ✓ *Work sampling*
- ✓ *Synthetic data*
- ✓ *Analytical estimating*
- ✓ *Predetermined Motion Time Study (PMTS)*

❖ **Note** - *Time study & Work sampling involve direct observation while remaining are data-based & analytical in nature*

WORK MEASUREMENT TECHNIQUES

<i>Techniques</i>	<i>Applications</i>	<i>Unit of measurements</i>
<i>Time study</i>	<i>Short cycle repetitive jobs. Widely used for direct work</i>	<i>Centi-minute (0.01 min)</i>
<i>Work sampling</i>	<i>Long cycle jobs</i>	<i>Minutes</i>
<i>Synthetic data</i>	<i>Short cycle repetitive jobs</i>	<i>Centi-minute</i>
<i>Analytical estimating</i>	<i>Short cycle non-repetitive jobs</i>	<i>Minutes</i>
<i>MTM</i>	<i>Manual operation confined to one work centre</i>	<i>TMU (1 TMU = 0.006min)</i>

TIME STUDY PROCEDURE



TIME STUDY – BASIC STEPS

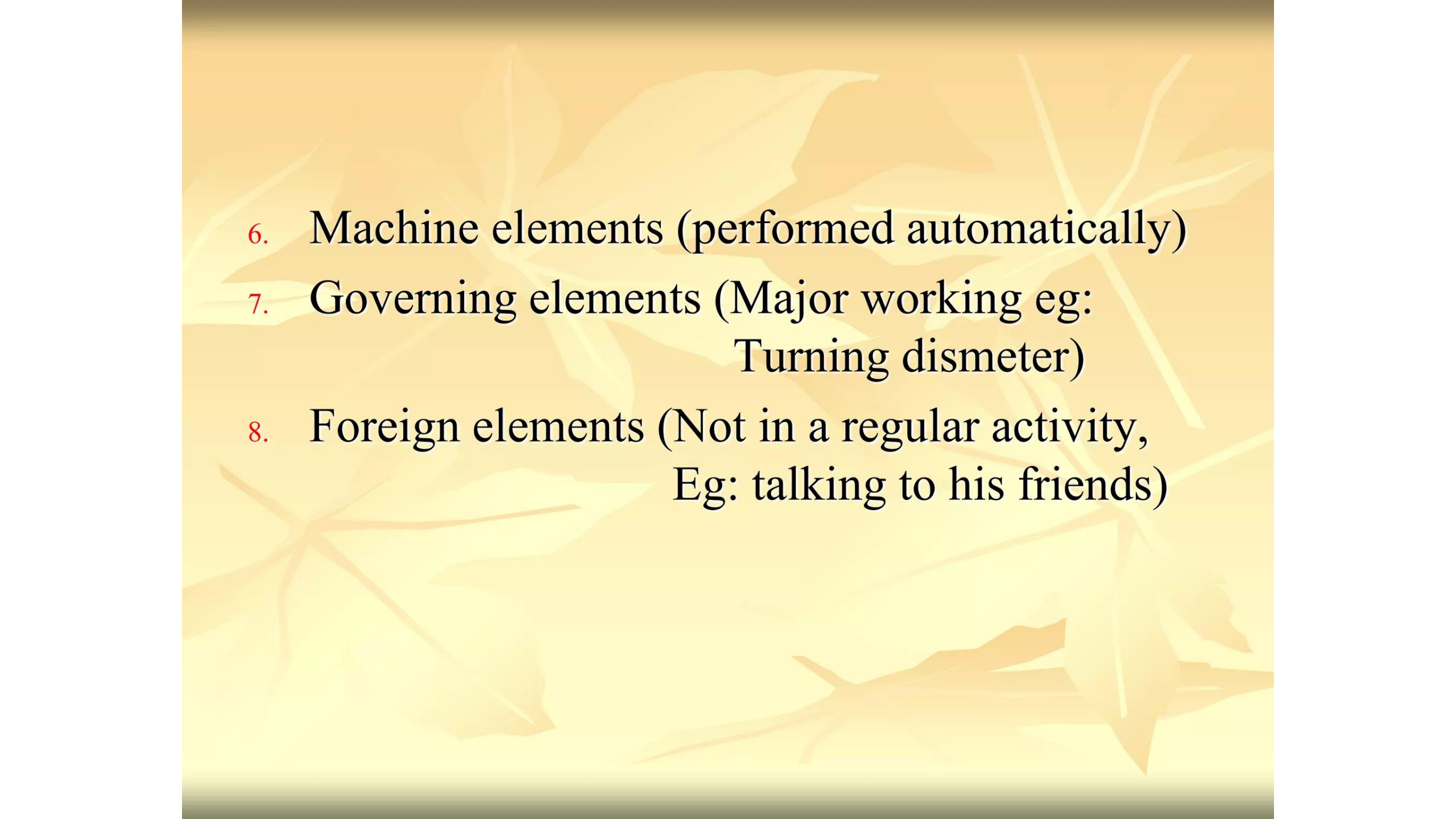
- a. Obtaining and recording all available information about the job, operator and the surrounding conditions likely to affect the execution of the work*
- b. Recording the complete description of the method, breaking down the operation into 'elements'*
- c. Measuring with a stopwatch and recording the time taken by the operator to perform each element of the operation*
- d. Assessing the rating*
- e. Extending observed time to 'basic times'*
- f. Determining the allowances to be made over and above the basic time for the operation*
- g. Determining the 'standard time' for the operation*

Breaking the job in to elements

- After collecting the data, breaking the job in to elements
- REASON
 - Ensure that production work is efficient
 - Better and accurate performance rating.
 - To identify different types of elements
 - To checking the method

Types of elements

1. Repetitive elements (Start, loading , stop the machine)
2. Occasional elements (Not doing every time
Eg: cleaning)
3. Constant elements (checking by using things ;
gauging diameters)
4. Variable elements (It varies not a constant)
5. Manual elements (Performed by worker)

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- 6. Machine elements (performed automatically)
 - 7. Governing elements (Major working eg:
Turning dismeter)
 - 8. Foreign elements (Not in a regular activity,
Eg: talking to his friends)

TIME STUDY EQUIPMENT

➤ *There are two methods of timing using a stop watch. They are:*

✓ *Fly back or Snap back method*

✓ *Continuous or Cumulative method*

TIME STUDY EQUIPMENT

1. Fly back Method:

Here the stop watch is started at the beginning of the first element. At the end of the element the reading is noted in the study sheet. At the same time, the stop watch hand is snapped back to zero. This is done by pressing down the knob, immediately the knob is released. The hand starts moving from zero for timing the next element. Thus the timing for each element found is called observed time.

2. Continuous method:

Here the stop watch is started at the beginning of the first element. The watch runs Continuously throughout the study. At the end of each element the watch readings are recorded on the study sheet. The time for each element is calculated by successive subtraction. The final reading of the stop watch gives the total time known as observed time.

TIME STUDY EQUIPMENT

➤ *Equipments used to measure time using Stop watch:*

- ✓ *Digital or electronics stop watch*
- ✓ *Electronic data collector and computer*
- ✓ *Observation board*
- ✓ *Observation sheet*
- ✓ *Stationary – Pen, Pencil, Eraser, Calculator*

Stop watch time study

- *It is a measurement technique*
- *Records the time and rates of working for the elements of a specified job.*
- ***TIME STUDY EQUIPMENT***
 - *Stop watch*
 - *Electronic devices*
 - *Motion picture camera*

■ STOP WATCH

- To measure the elemental timings
- It has two hands
 - i) Small hand (one revolution in 30 min)
 - ii) Large hand (One revolution in one minute)

■ TYPES OF STOPWATCH

- Non-fly back
- Fly back

- **Non-fly back (continuous)**

- First pressure on winding knob Starts the long hand
- Second pressure stops the long hand
- Third pressure returns hand to Zero position

- **Fly back**

- Start and stop with the help of slide
- Pressuring the wind knob brings the hand back to zero.

- **Non fly back:**

It is preferred for recording continuous timing.

- **Fly back:**

The watch is started and stopped with the help of slide.

- **Split hand:**

It gives higher accuracy in reading when two elements are to be timed successively.

